

Ashwin Krishnamurthy

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Mechatronics Engineer

EDUCATION

Incoming Freshman @ Carnegie Mellon University | Pittsburgh, PA
Weighted GPA: 4.0 / 4.0 | Mechanical Engineering & Planned Minor in Robotics | Class of 2030

High School Diploma @ Edison Academy Magnet School | Edison, NJ
Weighted GPA: 4.6115 / 4.0 | Mechanical/Civil Engineering Track | Class of 2026

AWARDS

NASA App Development Challenge Winner | February 2025
USA Physics Olympiad (USAPhO) Qualifier + Distinction (Top 400) | April 2025
AMC 10B Distinction (Top 5%) & 2x AIME Qualifier | November 2022, 2023
3rd Place Rutgers Malcolm G. McLaren Symposium | April 2025

SKILLS

Mechanical Engineering: Mechanisms, Statics, Materials Science, CAD
Mathematics: Calculus, Multivariable, Linear Algebra
Physics: Vehicle/Flight Dynamics, Astrophysics, Fluid Dynamics
Programming/Software: Python, C++, Java, Machine Learning, Unreal Engine, Blender

WORK EXPERIENCE

microMechanics of Deformation Research Group Intern @ Rutgers MSE | December 2024 - Present (In-Person Summer 2025)

- Novel analysis of **7304 minimum energy AI grain boundaries** spanning five macroscopic DoFs to develop classification theory

NASA STEM Enhancement in Earth Science (SEES) Research Intern | May 2025 - July 2025

- Selected for program (~5% Acc. rate) and performed **photometric analysis of NGC 2194** to determine and age and distance
- Presented poster at MIT Undergraduate Research Technology Conference and **American Geophysical Union** meeting

New Jersey Astronomical Association Qualified Observer | May 2024 - Present

- Certified **operator of 26-inch telescope** for public showcase and research @ NJAA Observatory
- Hosted booths at star parties, recruiting 100+ members for observatory, and involved in astrophotography and research groups

PROJECTS

NASA ADC | Co-Captain | Edison Academy Magnet School | September 2024 - Present

- Unreal Engine 5 simulation depicting the path of the Artemis II mission with key flight data
- Earned a **fully-funded trip to the Johnson Space Center** for a tour of internal facilities and a presentation to NASA executives
- Invited to speak at the **HELIX Discovery Showcase** to recognize the opening of the HELIX research building @ Rutgers

CubeSat | Co-Captain | MIT BWSI | December 2024 - Present

- Placed 4th** nationally in the MIT BWSI BACC for building a CubeSat to detect wildfires
- Working with mentors to **launch the CubeSat** via a weather balloon with a custom-made drone to capture footage

LEADERSHIP EXPERIENCE

Math League | Executive Board | Edison Academy Magnet School | June 2023 - Present

- Coordinate MCAMC** by securing sponsorships and writing problems. **Lead coach at Central Jersey Math Circle**; raised \$3k+ for ARML trip.

WeMake 4H | Executive Board | September 2022 - Present

- 150+ volunteer hours, 50+ classes**, and presentations @ 4H Fair and NJ Makers Day

Team Mechanical Lead | FIRST Tech Challenge Robotics Competition Team 24536 | October 2023 - Present

- Designed and constructed mechanisms and installed all sensors, earning us the **Innovate Award**